Section A — SQL Basics Documentation

This Section focuses on understanding the basic concepts of SQL using the ShopEase database.

The main objective of this section is to practice:

- Data retrieval using SELECT statements
- Viewing table structure
- Understanding Primary Keys
- Understanding database constraints such as UNIQUE, NOT NULL, and CHECK

This section builds the foundation for advanced SQL concepts in later sections.

Database Selection

Query

USE shopease_db;

Explanation

This command selects the database in which all queries will be executed.

Before working with tables, it is necessary to choose the correct database.

Q1. Display all columns and rows from customers table

Query

SELECT *

FROM customers;

Explanation

This query retrieves all columns and all rows from the customers table.

The * symbol means all available columns.

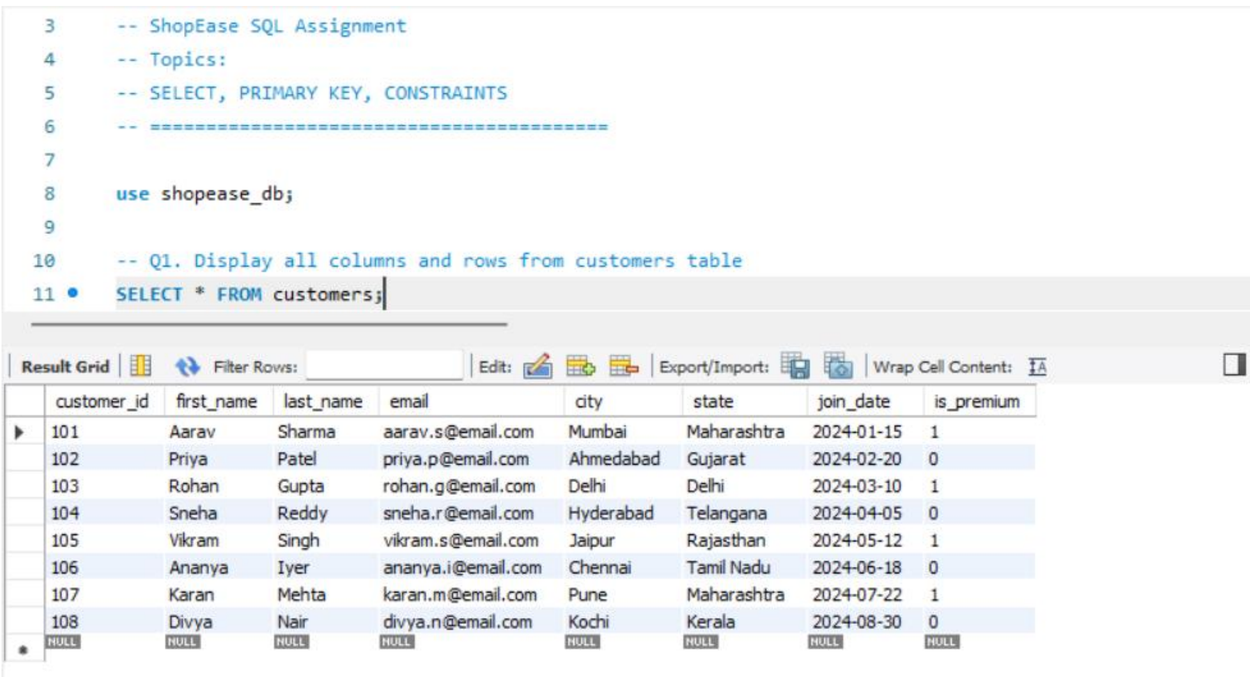
This helps in viewing complete customer information.

Expected Output

Displays complete customer details such as:

- customer_id
- first_name
- last_name
- email
- city
- state
- join_date
- is_premium

Screenshot



```
-- ShopEase SQL Assignment
-- Topics:
-- SELECT, PRIMARY KEY, CONSTRAINTS
-- ==========================================

use shopease_db;

-- Q1. Display all columns and rows from customers table
SELECT * FROM customers;
```


customer_id	first_name	last_name	email	city	state	join_date	is_premium
101	Aarav	Sharma	aarav.s@email.com	Mumbai	Maharashtra	2024-01-15	1
102	Priya	Patel	priya.p@email.com	Ahmedabad	Gujarat	2024-02-20	0
103	Rohan	Gupta	rohan.g@email.com	Delhi	Delhi	2024-03-10	1
104	Sneha	Reddy	sneha.r@email.com	Hyderabad	Telangana	2024-04-05	0
105	Vikram	Singh	vikram.s@email.com	Jaipur	Rajasthan	2024-05-12	1
106	Ananya	Iyer	ananya.i@email.com	Chennai	Tamil Nadu	2024-06-18	0
107	Karan	Mehta	karan.m@email.com	Pune	Maharashtra	2024-07-22	1
108	Divya	Nair	divya.n@email.com	Kochi	Kerala	2024-08-30	0
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Q2. Retrieve only first_name, last_name, and city of all customers

Query

```
SELECT
    first_name,
    last_name,
```

city

FROM customers;

Explanation

This query selects only specific columns from the customers table.

Instead of retrieving all data, it fetches only:

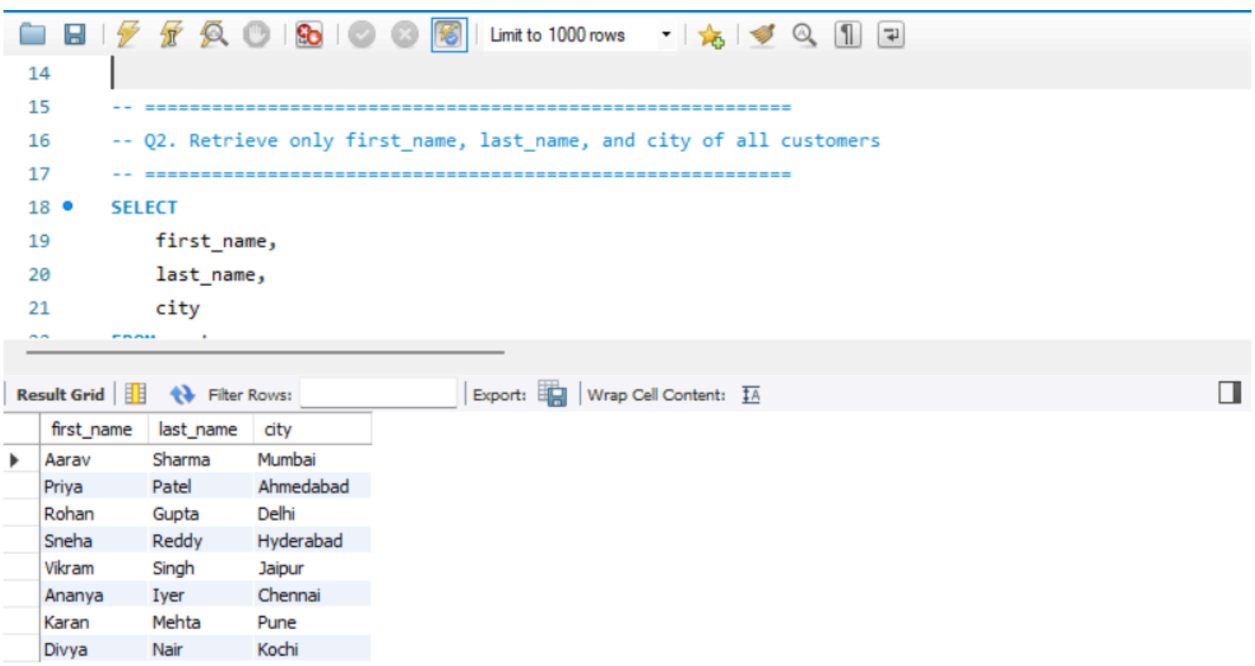
- first_name
- last_name
- city

This improves readability and reduces unnecessary output.

Expected Output

Displays customer names along with their city.

Screenshot



Q3. List all unique categories in products table

Query

SELECT DISTINCT category

FROM products;

Explanation

The DISTINCT keyword removes duplicate values and shows only unique categories.

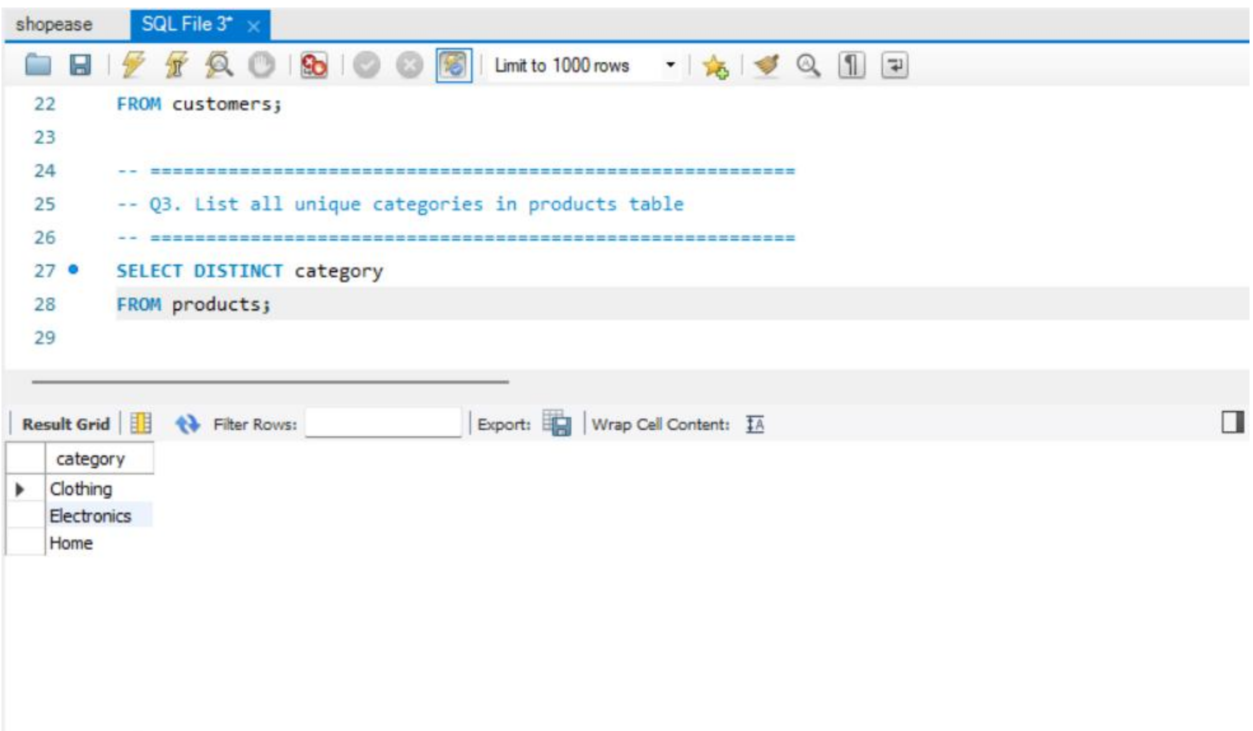
If multiple products belong to the same category, that category appears only once.

Expected Output

Example:

- Electronics
- Clothing
- Home

Screenshot



Q4. Identify Primary Key of each table

Query

DESCRIBE customers;

DESCRIBE products;

DESCRIBE orders;

DESCRIBE order_items;

Explanation

The DESCRIBE statement displays the structure of each table.

It shows:

- Column names
- Data types
- Constraints
- Keys

Primary Keys identified:

- customers → customer_id
- products → product_id
- orders → order_id
- order_items → item_id

A Primary Key must be:

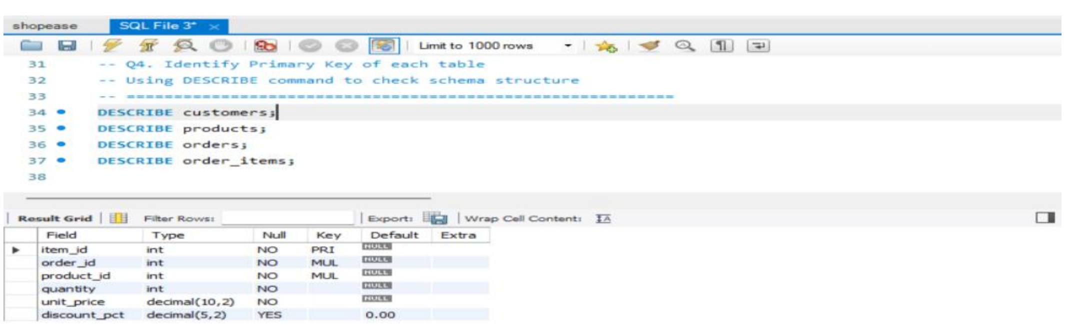
- Unique
- Not NULL

because it uniquely identifies each record.

Expected Output

Table schema with primary key details.

Screenshot



Q4 Practical Example — Duplicate Primary Key

Query

INSERT INTO customers VALUES

(101, 'Test', 'User', 'test@email.com', 'Pune', 'Maharashtra', '2024-01-01', TRUE);

Explanation

This query attempts to insert a duplicate customer_id.

Since customer_id is the Primary Key and already exists, the database rejects it.

Error

Duplicate entry '101' for key 'PRIMARY'

Q5. Email Constraints

Query

INSERT INTO customers VALUES

(109, 'Rahul', 'Patil', 'aarav.s@email.com', 'Nagpur', 'Maharashtra', '2024-09-01', FALSE);

Explanation

The email field has:

- UNIQUE constraint
- NOT NULL constraint

This query tries to insert an already existing email.

The database rejects it because duplicate emails are not allowed.

Error

Duplicate entry for email.

Q6. Insert product with negative unit_price

Query

INSERT INTO products VALUES

(209, 'Test Product', 'Electronics', 'TestBrand', -50, 100);

Explanation

The products table has a CHECK constraint:

unit_price > 0

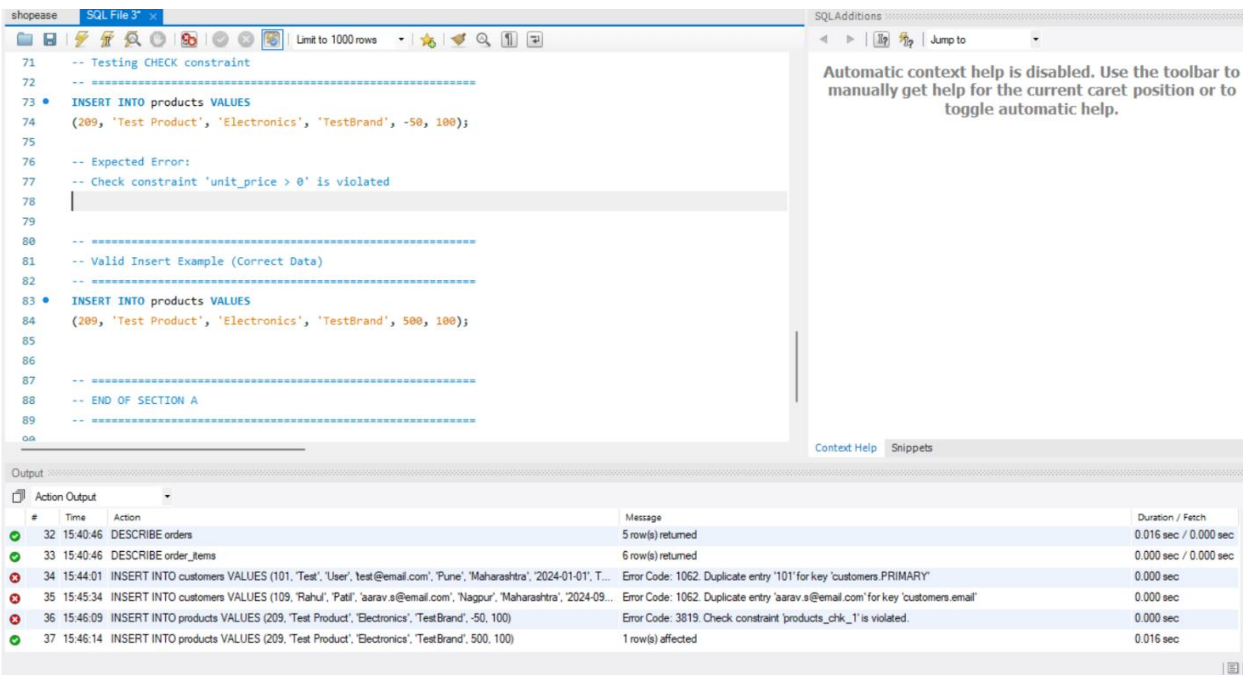
This query violates that rule by inserting a negative value.

The database prevents invalid data.

Error

Check constraint violated.

Screenshot



Valid Insert Example

Query

INSERT INTO products VALUES

(209, 'Test Product', 'Electronics', 'TestBrand', 500, 100);

Explanation

This query follows all constraints correctly.

The record is inserted successfully.

Screenshot

Concepts Covered

- SELECT
- DISTINCT
- DESCRIBE
- Primary Key
- UNIQUE Constraint
- NOT NULL Constraint
- CHECK Constraint
- Data Integrity

Summary

In this section, the basic SQL concepts were practiced using the ShopEase database.

The queries helped in understanding how data is retrieved, how table structures are inspected, and how constraints ensure data accuracy and consistency.